MomentsHub: Activity Diagram Analysis

Software Engineering - BCSE301P

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Assignment: Activity Diagram Documentation

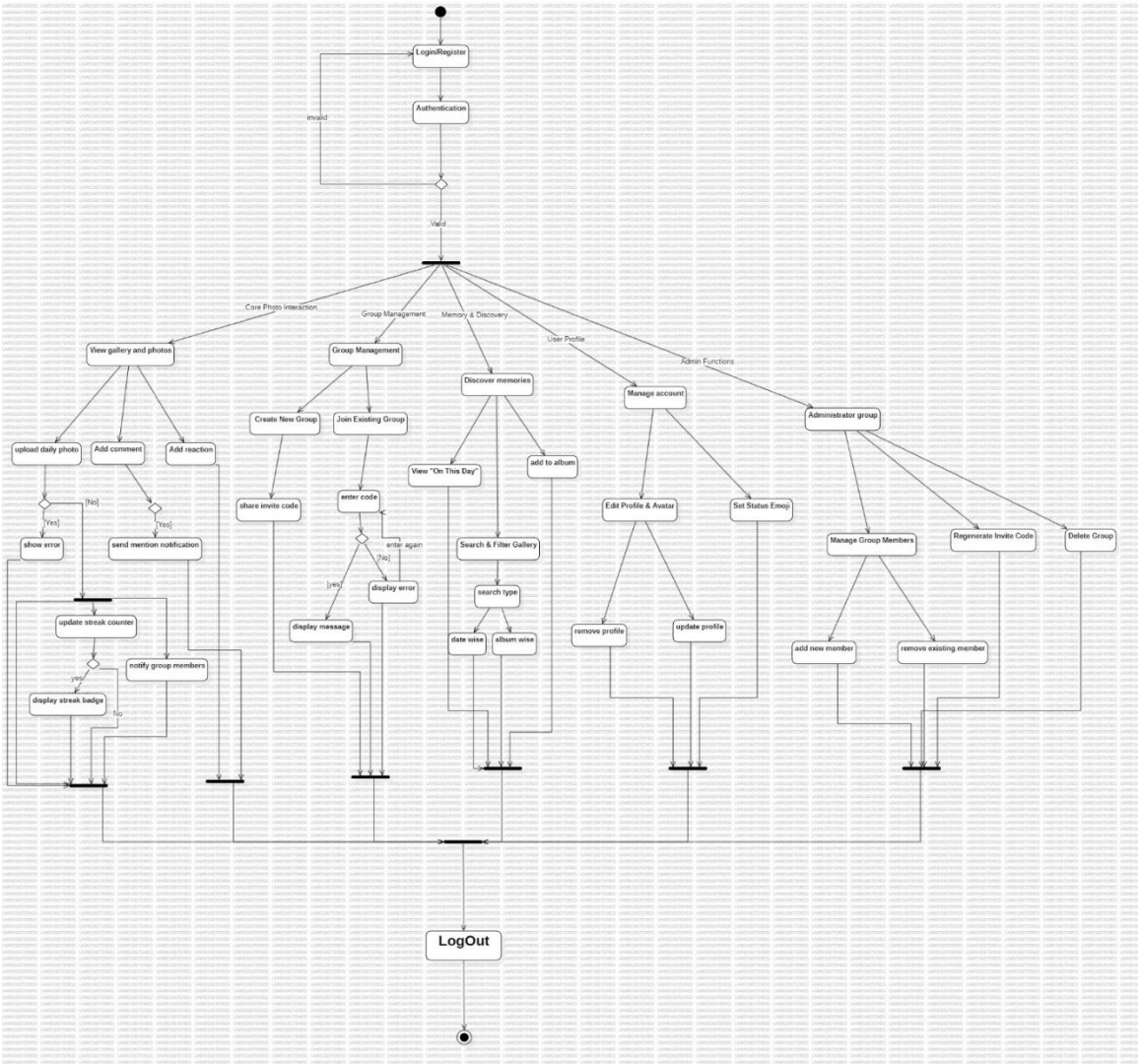
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1. Introduction

This document analyzes the Activity Diagram for MomentsHub: A Collaborative Daily Photo Capsule. The diagram illustrates the dynamic workflow of user interactions, system processes, and decision points within the application, covering authentication, photo sharing, group management, and discovery features.

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2. Key Workflows Analysis



2.1 Authentication Workflow

Flow: User accesses application → Login/Register decision → Firebase Authentication → Dashboard/Error handling

Key Activities:

- New user registration with email/password (FR1.1)
- Existing user authentication (FR1.2)
- Error handling for invalid credentials

Decision Points: Valid credentials check, account existence verification

2.2 Core Photo Interaction Workflow

Flow: Upload daily photo → Add caption → Gallery display → User interactions (comments, reactions, mentions)

Key Activities:

- One photo upload per day per group with mandatory caption (FR2.1, FR2.2)
- Emoji reactions (❤️ , 😂 , 🎉 , 🙏 , 🥰) and comment system (FR2.4, FR2.5)
- @mention functionality with notifications (FR2.6, FR2.7)

Decision Points: Photo validation (format, size, daily limit), mention detection in comments

2.3 Group Management Workflow

Flow: Create/Join group → Generate/Validate invite code → Group operations → Member management

Key Activities:

- Private group creation with admin role assignment (FR1.3)
- Unique invite code generation and validation (FR1.4, FR1.5)
- Admin functions: add/remove members, regenerate codes, delete groups

Decision Points: Admin permission verification, invite code validity, membership limits

2.4 Memory & Discovery Workflow

Flow: Access gallery → Apply filters (date/user/tags) → View "On This Day" → Browse results

Key Activities:

- "On This Day" feature displaying historical photos (FR3.1)
- Gallery search and filtering by date range, user, and tags (FR3.2-FR3.5)

Decision Points: Date-based content availability, filter criteria validation

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3. Activity Diagram Components

3.1 UML Elements Used

- **Start/End Nodes:** Single black circle (start), multiple end states for different completion scenarios
- **Activity Nodes:** User actions (upload, comment) and system processes (authentication, validation)
- **Decision Nodes:** Diamond shapes for conditional logic (authentication success/failure, permission checks)
- **Fork/Join Nodes:** Parallel activities like simultaneous photo processing and notification sending

3.2 System Integration Points

- **Firebase Authentication:** Secure login/registration handling
- **Firestore Database:** Real-time data storage and synchronization
- **Firebase Storage:** Photo file management with cloud storage
- **Real-time Listeners:** Live notifications and updates (2-second delivery target)

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4. Error Handling and Performance

4.1 Exception Flows

- **Authentication Errors:** Invalid credentials, network issues, service unavailability
- **Upload Failures:** Network timeout, file validation errors, daily limit exceeded
- **Group Management Errors:** Invalid invite codes, permission denied, capacity limits

4.2 Performance Integration

- Gallery loading under 3 seconds with thumbnail optimization

- Photo uploads completing within 10 seconds for 5MB files
- Real-time notifications delivered within 2 seconds

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5. Security Workflow

The diagram incorporates security measures:

- HTTPS encryption for all data transmission
- Firebase-managed password hashing and session handling
- Group membership validation and permission-based access control
- Firestore security rules enforcement for data access

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6. Development Implications

6.1 Frontend Development

- React component architecture aligned with activity flows
- State management for user sessions and real-time updates
- Form validation and error handling patterns

6.2 Backend Integration

- Firebase service integration points clearly defined
- API endpoint design following workflow patterns
- Database schema optimized for activity sequences

6.3 Testing Strategy

- Unit tests for individual activities and decision points
- Integration tests for complete workflow sequences
- Error scenario testing based on exception flows

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7. Conclusion

Creating this activity diagram helped me better understand how all the different parts of MomentsHub work together. The diagram shows the complete flow from when a user first opens the app to sharing photos with friends and family.

Through this exercise, I learned how to properly use UML activity diagram symbols like decision nodes and parallel activities. The diagram clearly maps to our functional requirements from the SRS, which will make it much easier when we start coding the actual application.

This activity diagram will definitely be useful as a reference when my teammate and I start building MomentsHub. It gives us a clear roadmap of what needs to be implemented in each iteration, and helps us understand how Firebase will handle the backend processes.